

Do heat pumps help integrate wind?

Sector coupling and transmission

Reading. Ruhnau et al. (2020), with the lecture note *Energy System Models* (ESM below), sections 3 and 5, as the toolbox. Section, figure, equation, table and footnote numbers are the paper's own. The paper's figures are not reprinted here, so keep it open at them.

Part 1 · Reading Ruhnau et al.

30 min

What heat pumps cost, and what they do for wind

Ruhnau et al. (2020) ask whether electrifying heat helps a power system absorb wind. Their tool is EMMA, a linear model of investment and hourly dispatch for five north-western European markets. The heat pumps are individual ones in buildings, and in this part they follow the heat load: they consume when heat is needed, whatever the price. This part reads §1–3.2 and Figs. 1–7.

1. The research question (§1).

- (i) The paper states three research questions (p. 2). Explain in a few sentences why they matter for the green transition.
- (ii) The paper calls the effect of heat pumps on wind's market value "ambiguous ex ante" (p. 2). What are the two opposing forces that make it so?

2. Heat pumps alone (§3.1, Figs. 1–3). How do wind power and heat pumps, each on its own, affect the system? In EMMA the rest of the system adjusts to the shares of wind and heat pumps the authors set: the model builds the other generation capacity, and it can shed load.

- (i) Explain Fig. 1.
- (ii) Investment in new capacity (Fig. 2): what type of capacity grows, by how much, and why?
- (iii) Load shedding (Figs. 2–3): what is the load shedding option, why is it used so much more at a 15% heat pump share, and why does its assumed cost barely move the load cost (Fig. 3, right)?
Hint: in a long-run equilibrium, what must the scarcity prices pay for, and how many hours do they need to do it?

3. Heat pumps and wind together (§3.2, Figs. 5–7).

- (i) Fig. 5: compare the daily and the seasonal patterns of wind and of the heat pump load. What does an hourly correlation of 0.11 (p. 6) say about the hope the introduction starts from?
- (ii) Fig. 6: what does wind do to the heat pump load cost, at low and at high heat pump shares? What do heat pumps do to wind's market value?
- (iii) Fig. 7: what does the dashed line hold fixed? Name the two effects of heat pumps on wind's value that the figure separates, and say which wins at 5–20% wind and which at 30%.

Hint: follow the amount of wind, in TWh, along the solid line and along the dashed one as heat pumps are added.

Stop point

The load cost is the price weighted by the heat pumps' consumption, the mirror of wind's market value. It rises steeply as heat pumps grow (by some 21 EUR/MWh from nothing to 15%), because a load that follows the heat demand consumes in the scarce hours whose prices pay for the peaking plant built to serve it. Heat pumps raise wind's value at a fixed wind capacity; at a fixed wind share, the extra wind takes the gain back, entirely at 30%.

If you finish early

Let λ_h be the price in hour h of a year of H hours, $\bar{\lambda}$ its average (the base price), and l_h the heat pump load, with average $\bar{l}$. Show that the load cost of Eq. (5) can be written

$$\text{load cost} = \bar{\lambda} + \frac{\text{Cov}(\lambda, l)}{\bar{l}},$$

with the covariance taken over the hours of the year. Divide by the base price to get a "cost factor", the value factor of ESM §3.4 turned around, and write it as one plus a product of a correlation and two coefficients of variation. Use it to explain Fig. 4 (right), including 2008 (footnote 12). What would a heat pump load that is uncorrelated with the price cost?

What takes the cost back, and what it does for wind

Sections 3.3–5 of the paper ask whether the load cost increase of part 1 can be brought down, and what that does for wind. The paper tries two remedies besides wind, "system-friendly" heat pumps (§3.3) and thermal storage (§3.4), and adds them up in §4.

What you need. For heat pumps that follow the heat load, the load cost rises steeply with their share, because they consume in the scarcest hours (from about 72 to 93 EUR/MWh at 15%, Fig. 1); wind takes part of that back (about 6 EUR/MWh at 30% wind, Fig. 6).

1. System-friendly heat pumps (§3.3, Fig. 8).

- (i) What makes a heat pump "system-friendly", and what does that mean for its electricity load?
- (ii) Fig. 8: what does a system-friendly heat pump do to the load cost, and why does it do little for wind's market value (p. 8)?

Hint: which hours does a steadier load stay out of, and how much does wind produce in them?

2. Flexible heat pumps (§3.4, Figs. 9–12).

- (i) What does the "flexibilization" of heat pumps with thermal storage mean, and how far can it go?
- (ii) Fig. 9: what does it do to the load cost? Figs. 10–12: why does it do little for wind's market value?

Hint: what pays for an interconnector or a pumped hydro plant?

3. What it adds up to (§4–5, Fig. 13).

- (i) Explain Fig. 13. What do the authors mean by saying the remedies can “overcompensate” the load cost increase (p. 11)?
- (ii) Sum up the paper’s answer to the question in this worksheet’s title: do heat pumps help integrate wind, or does wind help integrate heat pumps?

Stop point

System-friendly heat pumps draw a smaller and steadier load, and flexible ones move their load by a few hours. Both lower the load cost, and together with wind they more than offset its increase. Neither does much for wind’s value: a steadier load stays out of hours in which wind produces little, and storage crowds out interconnectors and pumped hydro. Wind helps heat pumps more than heat pumps help wind.

If you finish early

Eq. (9) with the paper’s losses: 5% of the heat put in is lost on the way in, and 1% of the heat stored is lost every hour. What share of a MWh of heat put into the storage can be taken out 4 hours later, a day later, a week later? If the heat pump’s coefficient of performance were the same in both hours, how much cheaper must electricity be in the hour the storage is charged, compared with the hour it is emptied, for the storage to be worth using in each case? What does that say about using this storage against the seasonal pattern of Fig. 5?